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Exploratory analysis of young adults' trajectories through the UK driving license acquisition process

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ABSTRACT

Objective: The UK is one of many high-income countries to experience a decline in driving license acquisition among young adults in the 2000s. This paper draws on newly available nationally representative microdata that captures the progress of individual drivers through the UK driving license acquisition process, to establish socio-demographic correlates.

Methods: Using the 2016 and 2017 editions of England's National Travel Survey data, a series of binary logit models were employed to identify factors associated with progression through the various phases of the UK's driving license acquisition process. Factors that are associated with (1) the frequency of taking the driving license tests, (2) the number of times having failed the theory and driving tests are then identified.

Results: The socio-demographic explanators considered were each found to be associated with driving license holding in intuitive ways that are consistent with prior literature. However, relatively few factors are significantly associated with progress through the steps of the license acquisition process, and the goodness-of-fit for progress through these intermediate phases are generally lower (indicating that other unobservable idiosyncratic personal or contextual characteristics are dominant in these processes). A consistent theme is the strong relationship with labor market participation. Links between income and the intermediate phases, however, were generally weaker. Age is negatively associated with progress through the early phases when respondents are applying for provisional license and taking theory test, but this relationship turns positive in later stages of the acquisition process.

Conclusion: To the authors' knowledge, this study is the first opportunity to evaluate this novel data resource covering the UK's driving license acquisition process. This is an important research direction to help policymakers understand young adults' delay in acquiring licenses, particularly the extent to which there may be structural inequalities. The main finding is that socio-demographic factors appear to be relatively poor predictors, with employment status the strongest single correlate of the variables that were tested. This paper is concluded with suggestions for designers of household travel surveys in regions where youth license-acquisition is of increasing focus, as well as a brief discussion of future research needs.

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Introduction

Young adults are delaying or forgoing acquiring a driving license in many industrialized countries, dating from the early 2000s (Kuhnimhof, Armoogum, et al. 2012; Delbosc and Currie 2013). The research question is why this phenomenon is occurring, with hypotheses including changing lifestyles, changing attitudes toward cars, and changing socio-economic status of young adults compared with earlier generations. For example, members of the current cohort of young adults tend to spend longer in education and to postpone marriage and parenthood (Office for National Statistics 2019a, 2019b). An increasing proportion live with their parents (Blumenberg et al. 2016). When compared with older generations, young adults nowadays are more likely to

live in densely populated urban areas which are well served by public transport (Delbosc and Nakanishi 2017), and these young adults are said to show a weak preference for cars but are open to any travel modes that are suitable (Delbosc and Nakanishi 2017). The slow rate of economic recovery since the Global Financial Crisis has also contributed to the decreasing automobile orientation (licensure, ownership, and use) among young adults (Ralph 2015; Williams 2017).

In contrast to these social/economic phenomena, a different set of explanations focus on changes in the process of acquiring a driving license. In England, the typical young adult in 2017 spends 1.5 years in the license-acquisition process (calculated by the authors from the UK National Travel Survey data [NTS]), which is comparable to findings from earlier research, including the 1.7 years in Le Vine and Polak

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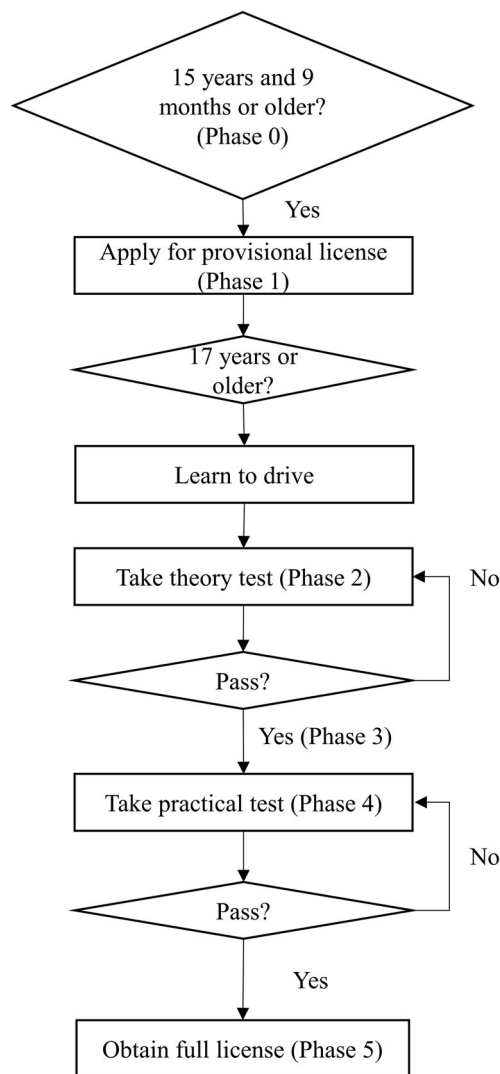


Figure 1. Driving license acquisition process in the UK and phases.

(2014) (data from year 2010 NTS) and the 13.2 months (excluding breaks of 6+ months) in the ‘Cohort II’ study (Wells et al. 2008) (survey data collected during years 2001–2005). Test pass rates have been stable since 2007 (the earliest available) for the “practical” (on-road) test (47% in 2007 vs 50% in 2019), however have trended downwards for the “theory” test (65% in 2007 vs 48% in 2019) (Department for Transport (DfT)) 2019). Though the license acquisition process varies across countries, a general observation in high income countries is a trend of increased stringency and complexity in the processes (see the Related Work section as well as this paper’s [Supplementary Material](#)).

The current driving license acquisition process in the UK is summarized in [Figure 1](#). There are three main milestones: acquiring a provisional license, passing the theory test, and then passing the driving test. With few exceptions, one can apply for a provisional license at 15 years and 9 months of age, but can only start driving after age 17. Candidates can take the theory test at any time (subject to test center availability), and can do the same for the driving test after they have passed the theory test. Passing the driving test leads immediately to acquisition of a full driving license.

For driving license acquisition, since its inception in 1965, the NTS only asked what type of license the respondent holds (no license, provisional license or full license). However, beginning in 2015 the NTS started to collect detailed driving license acquisition data through additional questions. These data allow researchers to investigate factors that are associated with the driving license acquisition process to greater depth than was previously possible. To the authors’ knowledge, this is the first time that NTS collects driving license acquisition data, and it is the first opportunity to evaluate the driving license acquisition process. Given the imperative of decarbonizing the transport sector, license-acquisition is an important step at which policy may be targeted. Our objective is to contribute to better understanding current license-acquisition behavior, which is a pre-requisite to informing the design of policies to change the status quo.

Related work

The main factors explored in prior work include socio-demographic features, the built environment, the possibility of changing attitudes of young adults to cars, and the introduction of graduated driving license (GDL) systems. This paper’s [Supplementary Material](#) maps and organizes the relevant literature by study objectives, geography, data sources, methodology, and factors.

The literature focuses primarily on industrialized western countries, especially the US (i.e., Brown and Handy (2015)) and Australia (i.e., Raimond and Milthorpe (2010)). Several studies focus on European countries, including the UK (Le Vine and Polak 2014), France (Bayart et al. 2020), and Norway (Hjorthol 2016). In part the under-representation of developing countries is related to the lack of statistically representative national travel surveys (see Le Vine et al. (2018)).

A general finding is that being employed and having high income increases the probability of holding a license (Le Vine and Polak 2014; Brown and Handy 2015), and Kuhnimhof, Wirtz, et al. (2012) report falling incomes of young adults as one correlate of declining automobile orientation. Living with parents and having good accessibility (from one’s residence) to public transport services decreases the probability of holding a license (Le Vine and Polak 2014; Bayart et al. 2020). Hence, the literature suggests that the delay in various milestones of adulthood, re-urbanization, and increasing financial pressure on young adults may be associated with the overall observation of delayed driving license acquisition.

Other factors explored include attitudes and the influence of Information and Communication Technologies (ICT). With respect to attitudes, some studies argue that the young generation see car ownership as less of a status symbol than earlier generations did, and instead as an indicator of maturity and independence (i.e., Fylan and Caveney (2018)). Thigpen and Handy (2018) reported mixed findings regarding the effect of one’s social network on the timing of license-acquisition. The authors also found that reporting “enjoyment” from cycling or public transport was associated

with delay. Multiple studies have found no evidence of heightened environmental concern being associated with delay in license acquisition (Delbosc and Currie 2014; Schoettle and Sivak 2014). With respect to ICT, Thigpen and Handy (2018) found ICT delays driving license acquisition, however other studies report that possessing a smartphone and/or using electronic communication *increase* the likelihood of young adults having a driving license (Brown and Handy 2015).

GDL has been introduced in many high-income countries, including Australia, Canada, New Zealand and the US [see reviews in Senserrick and Williams (2015) and Table SM2, as well as extended discussion in the [Supplementary Material](#)]. Although the UK's Department for Transport has recently studied GDL (Department for Transport 2019), it has not yet elected to introduce it due to stated concerns that it would potentially restrict young people's education and employment (Department for Transport 2019), GDL and the UK's multi-stage driving license acquisition system have similarities (see detailed comparisons in Section SM1). The overall influence of GDL on license acquisition is mixed: Raimond and Milthorpe (2010) and Tefft et al. (2014) did not find that GDL led to delayed license acquisition, however Thigpen and Handy (2018) found that it did.

Wells et al. (2008) reported findings from a major longitudinal study of young and novice British drivers. Key findings include younger age groups having a higher rate of test-passing, and females having a higher rate of passing the theory test whereas males have a higher pass rate for the driving test.

The present study builds on the earlier Wells et al. (2008) work by introducing multivariate regression analysis of individual drivers' progress through the license-acquisition regime, including a richer set of socio-demographic features than age and gender (which were analyzed by Wells and colleagues). The present study is motivated to better understand learner-drivers' trajectories up to the point of license acquisition. Results of this paper, which is based on a novel dataset, are also in contrast to many prior studies that have typically drawn on observations of the outcomes of these processes (license acquisition or not, at a given age).

Methods

This section first introduces the data resource employed in this study, followed by the descriptive analysis of the data and then the statistical approach.

Data

The questions newly added in 2015 that are asked to only provisional license holders, enquiring about their progress through the license acquisition process (Department for Transport 2017), are as follows:

- Are you currently learning to drive?
- [If currently learning to drive] Have you taken the driving theory test?

- [If you have taken theory test] When did you first take the driving theory test (year and month)?
- [If you have taken theory test] Have you passed the driving theory test?
- [If you have passed theory test] How many times did you fail the theory test before passing?
- [If you have passed theory test] Have you taken the practical driving test?
- [If you have passed theory test] When do you plan to [next] take the practical test?
- [If you have taken practical test] When did you first take the practical test (year and month)?
- [If you have taken practical test more than once] When did you most recently take the practical test (year and month)?

Figure 2 illustrates the relationship between the driving license acquisition process and the newly added process-related questions.

Descriptive analysis of license-acquisition-process data

The driving license acquisition process was coded into its five phases (see Table 1). The number of times failing the theory test and the number of times failing the practical driving tests was summed up, and this quantity was divided by the duration between the first time the respondent took the theory test and the date of the respondent's NTS

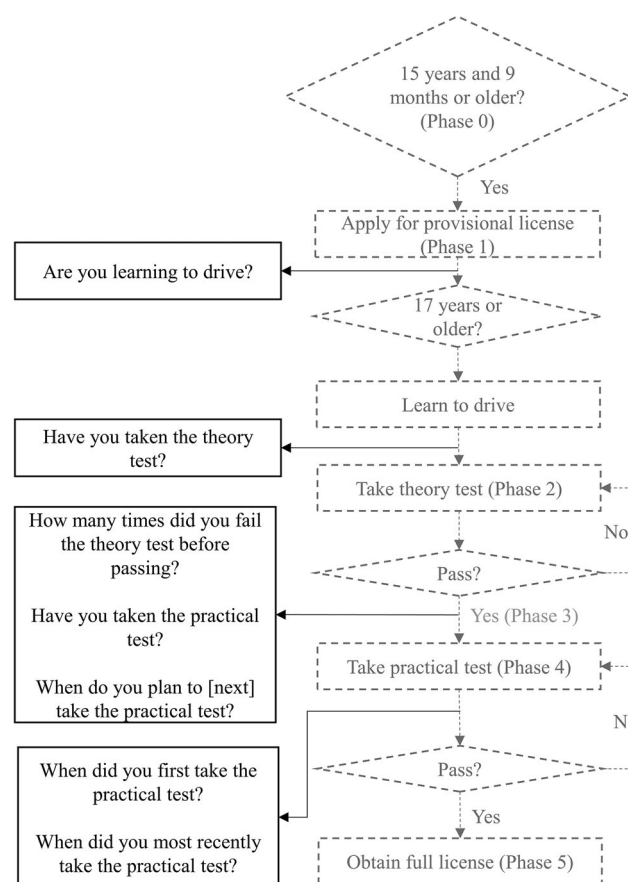


Figure 2. Relationship between steps of the driving license acquisition process (right) and NTS survey questions (left).

Table 1. Descriptive analysis of 2016–2017 adults.

Variable Name	Distribution or mean value
Phase: No license (Phase = 0)	19.6%
Phase: Provisional license, has not taken any tests (Phase = 1)	5.6%
Phase: Provisional license, has taken but not passed the theory tests (Phase = 2)	0.3%
Phase: Provisional license, has passed the theory tests (Phase = 3)	0.7%
Phase: Provisional license, has taken but not passed the driving tests (Phase = 4)	0.3%
Phase: Full license (Phase = 5)	73.5%
Test-taking frequency per year	3.0 times (N = 273)
Number of times having failed the theory tests	1.0 times (N = 273)
Number of times having failed the practical driving tests	2.3 times (N = 78)
Sample size	26,501

interview to calculate the frequency of taking license-acquisition tests per year (shown in Table 1).

Table SM3 of the **Supplementary Material** contains a summary of the socio-economic profiles of young British adults in both years 2016/17 (the focus of this analysis) and comparison with 2002/03, as 2002 is first year when the NTS sample was increased to its present size (along with various other structural changes to the survey protocol). Due to the enforced interregnum period between age 15 years, 9 months (when a candidate can apply for a provisional license) and when driving is allowed to commence at age 17 (see Figure 1), the analysis presented in this paper excludes all NTS respondents under age 17.

Although the NTS is overall a large-sample survey (N = 17,633 and 16,425 in 2016 and 2017, respectively), the relatively small proportion of the population that is actively in the process of acquiring a driving license means that this subsample is much smaller. Hence, numerical analysis on the number of tests taken are based on a sample size of N = 273 for the number of times having failed in the theory test and N = 78 for the number of times having failed the practical driving tests.

Statistical regression models

The statistical models employed in this study are binomial logit (BNL) and linear regression models. BNL was employed to analyze the correlates of socio-demographic features and (1) respondents having a full license; (2) respondents' progress in the license acquisition process, and (3) factors that are associated with multiple failings in the theory test and the practical driving test. Linear regression was used to analyze the associations of socio-demographic features and test taking frequency.

Results

Factors associated with young people's full driving license possession was analyzed first (Table SM4). Other than gender and walking time to the nearest public transit station, all

independent variables for the with/without full driving license model are highly significant. The signs of each of these parameters is the same as prior analysis performed using year 2010 data, which is shown in Table SM4 for comparison purposes (Le Vine and Polak 2014). Their *ceteris paribus* relationships with full driving license holding are:

- Age: positive association (0.16, $p < 0.01$)
- Income: positive for both individual (2.17×10^{-5} , $p < 0.01$) and other household members' income (1.16×10^{-5} , $p < 0.01$), and the individual income has higher impact than other household members' income
- Employment status (and full-time employed): positive association (0.74, $p < 0.01$ for employment status and 0.37, $p < 0.01$ for full-time employed)
- Degree level: positive association (0.41, $p < 0.01$)
- Born in the UK: positive association (0.56, $p < 0.01$)
- Population density: negative association (-0.01 , $p < 0.01$)
- Living in London and other urban areas: negative association (-0.72 , $p < 0.01$ for living in London and -0.45 , $p < 0.01$ for living in other urban areas)
- Living with adults a generation (16+ years) older: negative association (-0.40 , $p < 0.01$)
- Live with children a generation (16+ years) younger: negative association (-0.38 , $p < 0.01$).

Notable changes from year 2010 include gender becoming insignificant (i.e., being male has a $p = 0.47$ for year 2016/17 and $p < 0.01$ for year 2010), and living with children 16 or more years younger becoming significant ($p < 0.01$ for year 2016/17 and $p = 0.82$ for year 2010).

Table SM5 presents the first of the results based on the acquisition-process data; four BNL models are shown that capture respondents' progress from no license through to the practical driving test. The sample size decreases from left to right in Table SM5 (from 7,116 to 266), because in the model of each successive milestone of the license-acquisition process, respondents who have not entered the phase are excluded. For instance, the model for whether a respondent has or has not taken the theory test (Phase 1 versus Phases 2–4) does not consider people that have not yet acquired a provisional license (Phase 0).

The overall goodness-of-fit for the models presented in both Tables SM5 and SM6 are relatively low (five out of seven models have pseudo $r^2 < 0.1$). The general conclusion is therefore that standard socio-demographic characteristics provide relatively little explanatory power for progress through each of the phases of license-acquisition. In Table SM5, the higher goodness-of-fit for the first model (pseudo $r^2 = 0.28$) indicates that the socio-demographic variables considered provide better explanation for which young adults have initiated the license-acquisition process, in comparison to their progress in taking/passing the tests. Other explanators, which are unobservable in the NTS data, appear to be more salient.

Across the four models presented in Table SM5, age and employment status tend to be the strongest predictors, each significant (with p -value < 0.06) in three of the four models. Being employed is consistently a positive effect, however

individual income and full-time employment status are weaker predictors (significant in only one of the four models).

The effect of age is negative at the early stages of the license acquisition process (-0.05 and -0.04 in the first and second models, respectively), but it trends in a positive direction in the final phase (having taken the driving test, with coefficient = 0.05). Thus the overall positive *ceteris paribus* relationship between age and full-license-holding status (see Table SM4) is primarily due to older young adults progressing more rapidly through the later phases of the process (rather than the earlier phases). For the model of whether a candidate has taken the practical driving test (and failed, because they would become fully licensed by passing), the parameter for age is the only significant effect.

Having been born in the UK is significant in the models of the first two phases, but its influence is mixed: It has a positive impact on applying for a provisional license (0.23 , $p < 0.01$), but a negative impact (-0.84 , $p < 0.01$) on starting to take theory test. In other words, international migrants to the UK tend, all else equal, to move more quickly to taking the theory test rather than remain in Phase 0 (holding a provisional license but not having taken any tests).

Table SM6 presents results from the frequency of test-taking (tests taken/year), the number of times having failed the theory test, and the number of times having failed the practical driving test.

A model of the relationship between the frequency of test-taking per year and the socio-demographic features is presented first. The logic is that licensure candidates that engage in test-taking at a high frequency are likely to be relatively highly motivated to obtain a driving license and/or experiencing difficulty passing the tests (the NTS data does not allow us to distinguish between these competing explanations).

Consistent with the results previously presented from the models of progress through the phases of license acquisition, the results for this model of test-taking frequency show a positive and significant association between test frequency and employment status (1.23 , $p = 0.07$). Individual income is not significant, whereas other household members' income (parental income in the case of many young adults) has a positive association (3.45×10^{-5} , $p = 0.01$) with the frequency of test-taking.

The other two models in Table SM6 are binary logit models which analyze factors associated with experiencing multiple (two or more) test failures. Similar to the linear model, few significant relationships between the dependent variables and with socio-demographic factors were found. However, employment status again shows significant and positive association with multiple failings in both theory (is employed, with coefficient = 0.93 and $p = 0.07$) and driving (is full-time employed, with coefficient = 0.84 and $p = 0.11$) tests. The impact of gender on the two tests is opposite: being male is associated, net of confounding effects, with higher numbers of failures of the theory test (0.80 and $p = 0.07$), but vice versa for the driving test (-0.99 and $p = 0.10$). Being born in the UK is associated with lower numbers of failures of the theory test, which may be because

they are more familiar with the traffic rules in the UK (-0.95 and $p = 0.06$). Candidates living in low-density locations are weakly significantly (1.07×10^{-2} , $p = 0.07$) less likely to have multiple failures of the driving test, though this effect was not found for the theory test.

Discussion

This study contributes an initial exploratory analysis of young adults' progress through the elemental steps of the UK's driving license acquisition process. Earlier work identified that the typical British young adult is (as of 2010) spending between one and two years in the process of acquiring a driving license, which was found continues to be the case in 2016/17 (at 1.5 years).

There are competing hypotheses as to whether this is due to the increasing complexity of license-acquisition regimes or changing preferences. To the best of the authors' knowledge, this is the first study to evaluate these changes using acquisition-process data of this level of granularity. However, while the National Travel Survey dataset that was employed is large-sample and nationally-representative of the English population in 2016/17, an important limitation of this study is the small sample size of the number of individuals observed in each phase of the license-acquisition process.

The analysis focuses on two main questions: (1) which factors are associated with progress through the phases from being unlicensed to holding a full driving license; and (2) which factors contribute to experiencing multiple failed tests and high rates of test-taking (with the latter measured as frequency of test-taking/year).

Many standard socio-demographic explanators were not found significantly to correlate with progress through the process of acquiring a driving license, though they are better able to predict which young adults hold a full driving license (i.e., which have successfully completed all steps in the process). Goodness-of-fit is generally low, indicating that other unobservable idiosyncratic personal or contextual characteristics are dominant in these processes. A consistent theme through many of the models of progress through the individual phases is the structural importance of labor market participation, with much weaker patterns for other socio-demographic indicators. Employment status was also found to correlate, *ceteris paribus*, with test-taking frequency. A mixed and generally weaker patterns for income is found, though in several models income of other household members (which will include parental income of candidates living with parents) is positively related to progress through the license-acquisition regime as well as frequency of test-taking.

Beyond the small-sample limitations noted previously, this line of research highlights the complexity of gathering data of progress through the license-acquisition-process in a large-scale household travel survey. In future surveys, it would be very useful to collect process-related data from people that have recently completed the process, which is not observable using the English NTS data resource and greatly limits researchers' ability to distinguish between the trajectories of the most successful learner-drivers and others. It would be

useful to add several additional data points that are not present in the NTS data which would permit alternative statistical approaches (e.g., inclusion of additional detail about the timing of the transition between each phase would permit use of survival modeling approaches). It would also be desirable to include questions regarding the reasons for the amount of passage of time by each individual candidate in the license acquisition process. Data that were employed in this study do not indicate whether the passage of time is voluntary on the part of the license-seeker or externally imposed (perhaps because of demand for test-taking in excess of available test-taking slots). The limited saliency of socio-demographic explanators suggests that it may be useful to gather more information from respondents about their behavior and decisions whilst learning to drive. Qualitative research methods could be very helpful to better understand which contextual or personal factors that are unobserved in this study are important in explaining successful navigation of the process. International comparisons from other locations that have experienced declines in youth licensing would help advance this line of research, as would further enquiry on the strong link between labor market participation and license-acquisition, specifically on the dual transitions of young adults into both employment and into driving.

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